Python Basics



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Content

- Comments in Python
- Variable Declaration in Python
- Data Types in Python
- Type Conversion in Python
- Operators in Python

Comments in Python

- In general, Comments are used in a programming language to describe the program or to hide the some part of code from the interpreter.

- Comments in Python can be used to explain any program code. It can also be used to hide the code as well.

- Comment is not a part of the program, but it enhances the interactivity of the program and makes the program readable.

Python supports two types of comments:

- Single Line Comment
- Multi Line Comment

Comments in Python

Cont..

Single Line Comment:

In case user wants to specify a single line comment, then comment must start with **'#'**

Example:

```
# This is single line comment
print "Hello Python"
```

Output:

Hello Python

Multi Line Comment:

Multi lined comment can be given inside triple quotes.

Example:

```
'''This is
   Multiline
   Comment'''
print "Hello Python"
```

Output:

Hello Python

Variable Declaration in Python

- A variable is a named memory location in which we can store values for the particular program.

- In other words, Variable is a name which is used to refer memory location. Variable also known as identifier and used to hold value.

- In Python, We don't need to declare explicitly variable in Python. When we assign any value to the variable that variable is declared automatically.

- In Python, We don't need to specify the type of variable because Python is a loosely typed language.

Variable Declaration in Python

Cont..

- In loosely typed language no need to specify the type of variable because the variable automatically changes it's data type based on assigned value.

Rules for naming variable:

- Variable names can be a group of both letters and digits, but they have to begin with a letter or an underscore.
- It is recommended to use lowercase letters for variable name. ‘SUM’ and ‘sum’ both are two different variables.

Example: Vardemo.py

```
a=10 #integer
b="StudyGlance" #string
c=12.5 #float
print(a)
print(b)
print(c)
```

output:

$python3 Vardemo.py

```
10
StudyGlance
12.5
```

Variable Declaration in Python

Cont..

- Python allows us to assign a value to multiple variables and multiple values to multiple variables in a single statement which is also known as multiple assignment.
- **Assign single value to multiple variables :**

Example: Vardemo1.py

```
x=y=z=50
print x
print y
print z
```

output:

```
$python3 Vardemo1.py
50
50
50
```

- **Assign multiple values to multiple variables :**

Example: Vardemo2.py

```
a,b,c=5,10,15
print a
print b
print c
```

output:

```
$python3 Vardemo2.py
5
10
15
```

Data Types in Python

- In general, Data Types specifies what type of data will be stored in variables. Variables can hold values of different data types.

- Python is a dynamically typed or loosely typed language, hence we need not define the type of the variable while declaring it.

- The interpreter implicitly binds the value with its type.

- Python provides us the **type ()** function which enables us to check the type of the variable.

Data Types in Python

Cont..

- Python provides following **standard data types**, those are
  - ✓ **Numbers**
  - ✓ **String**

Numbers:

- Number stores numeric values. Python creates Number type variable when a number is assigned to a variable.

There are three numeric types in Python:

1. int
2. float
3. Complex

Data Types in Python

Cont..

1. int:

Int, or integer, is a whole number, positive or negative, without decimals, of unlimited length.

Example:

a=10
b=-12
c=123456789

2. float:

Float or "floating point number" is a number, positive or negative, containing one or more decimals.

Example:

X=1.0
Y=12.3
Z=-13.4

3. complex:

Complex numbers are written with a "j" as the imaginary part.

Example:

A=2+5j
B=-3+4j
C=-6j

Data Types in Python

Cont..

String:

- The string can be defined as the sequence of characters represented in the quotation marks. In python, we can use single, double, or triple quotes to define a string.

- In the case of string handling, the operator + is used to concatenate two strings as the operation *"hello"+" python"* returns *"hello python"*.

Example:

```
S1='Welcome'  #using single quotes
S2="To"       #using double quotes
S3='''Python''' #using triple quotes
```

Data Types in Python

Cont..

Example: “datatypesdemo.py”

```
a=10
b=“Python"
c = 10.5
d=2.14j
print("Data type of Variable a :",type(a))
print("Data type of Variable b :",type(b))
print("Data type of Variable c :",type(c))
print("Data type of Variable d :",type(d))
```

Output:

python3 datatypesdemo.py

Datatype of Variable a : <class ‘int’>

Datatype of Variable b : <class ‘str’>

Datatype of Variable c : <class ‘float’>

Datatype of Variable d : <class ‘complex’>

Type Conversion in Python

- Python provides Explicit type conversion functions to directly convert one data type to another. It is also called as **Type Casting** in Python
- Python supports following functions
  1. **int ()** : This function converts **any data type to integer.**
  2. **float()** : This function is used to convert **any data type to a floating point number.**
  3. **str()** : This function is used to convert **any data type to a string.**

Example:"Typeconversiondemo.py"

```
x = int(2.8)
y = int("3")
z = float(2)
s = str(10)
print(x);print(y)
print(z); print(s)
```

Output:

```
python3 typeconversiondemo.py
2
3
2
10
```

Operators in Python

- The operator can be defined as a symbol which is responsible for a particular operation between two operands.
- Python provides a variety of operators described as follows.

Arithmetic operators :

+ (addition)	**eg:** a=20; b=10 then a **+** b=30
- (subtraction)	**eg:** a=20; b=10 then a **-** b=10
***(multiplication)	**eg:** a=20; b=10 then a * b=200
/ (divide)	**eg:** a=20; b=10 then a **/** b=2
%(reminder)	**eg:** a=20; b=10 then a **%** b=0
// (floor division)	**eg:** a=24; b=7 then a **//** b=3
** (exponent)	**eg:** a=2; b=3 then a ** b=8

Membership operators :

in (True, If the value is present in the data structure)

not in (True, If the value is not present in the data structure)

Operators in Python

Cont…

Comparison operators :

== (Equal to)
!= (Not equal to)
<= (Less than or equal)
>= (Greater than or equal)
< (Less than)
> (Greater than)

Bitwise operators :

& (binary and)
| (binary or)
^ (binary xor)
~ (negation)
<< (left shift)
>> (right shift)

Assignment operators :

= (Assigns to)
+= (Assignment after Addition)
-= (Assignment after Subtraction)
***=** (Assignment after Multiplication)
/= (Assignment after Division)
%= (Assignment after Modulus)
****=** (Assignment after Exponent)
//= (Assignment after floor division)

Logical operators :

and (logical and)
or (logical or)
not (logical not)

Identity operators :

is (Returns true if both variables are the same object)

is not (Returns true if both variables are not the same object)